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The C27n-C26r geomagnetic polarity reversal recorded in the west Greenland flood basalt province: How complex is the transitional field?

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[1] We present a paleomagnetic record of the C27n-C26r (~60.9 Ma) geomagnetic polarity reversal (GPR) sampled in two separate lava sequences in the Vaigat Formation, west Greenland flood basalt province. The two profiles, distanced 16 km apart, consist of 45 and 35 sampled lava flows of which 31 and 23, respectively, record transitional field directions. A prominent feature of each profile is the presence of successive flows with similar paleomagnetic directions, which based on geological, geochemical, and paleomagnetic evidences are interpreted to represent flow fields (i.e., lavas erupted in a short period of time) rather than periods when the geomagnetic field stood still. Multimodel photogrammetry shows that the typical flow length of Vaigat Formation picritic lavas is <2 km, which explains why flow-to-flow correlation of paleomagnetic directions between the two profiles (16 km apart) is not possible. The two profiles therefore record different and independent time series of the same GPR, allowing us to cast light on some of the questions that have arisen from single records. Particularly noteworthy is the fact that the two high-resolution paleomagnetic records share no general trends, clearly indicating that the C27n-C26r transitional field was more complex and faster changing than can be resolved by either record. In addition, we obtained stable polarity paleomagnetic data from both pretransitional C27n and posttransitional C26r yielding a paleomagnetic pole latitude 64.8°N, longitude 141.5°E ($N = 14$, $K = 19.8$, $A_{95} = 9.2°$) that passes reversal test. *INDEX TERMS:* 1224 Geodesy and Gravity: Photogrammetry; 1530 Geomagnetism and Paleomagnetism: Rapid time variations; 1535 Geomagnetism and Paleomagnetism: Reversals (process, timescale, magnetostratigraphy); 9604 Information Related to Geologic Time: Cenozoic; *KEYWORDS:* North Atlantic igneous province, Greenland, geomagnetic polarity reversal, paleomagnetic pole

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1. Introduction

[2] During the past 25 years, the paleomagnetic community has invested substantial effort in obtaining records of geomagnetic polarity reversals (GPRs). The main motivation for these studies is the belief that GPR records hold important information about the factors controlling the geodynamo and hence the physical conditions in the Earth's core and lower mantle. Recent advances in numerical geodynamo models further encourage such studies as the models become increasingly more realistic and, for example, different heat flux conditions at the core-mantle boundary can be tested against actual paleomagnetic data [*Glatzmaier et al.*, 1999; *Coe et al.*, 2000].

[3] Unfortunately, it has proven difficult to find geological sequences in which the relative short duration GPRs, ~1–8 kyr [*Merrill and McFadden*, 1999], have been reliably recorded in sufficient temporal detail. The geomagnetic significance of the 151 paleomagnetic GPR records listed in the IAGA Trans00 database [*McElhinny and Lock*, 1995] remains unclear, as demonstrated by the fact that different subsets have led scientists to draw widely different conclusions [*Merrill and McFadden*, 1999]. Particularly intriguing is the difference between the longitudinally confined virtual

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geomagnetic pole (VGP) paths observed in sedimentary sequences (~75% of the database records [*Laj et al.*, 1991; *Vizan and Van Zele*, 2001]) and the unconfined VGP paths from lava sequences [*Prévot and Camps*, 1993]. Several authors have suggested that this difference could reflect an artefact in sedimentary GPR records as paleomagnetic data from sediments are particularly problematic for weak and fast changing fields [*Rochette*, 1990; *Langereis et al.*, 1992; *Quidelleur et al.*, 1995]. Lavas, on the other hand, reliably record weak and rapidly changing GPR fields. Analyses of volcanic GPRs are, however, complicated by the discontinuous nature of volcanism and different conclusions regarding longitudinal confinement have been reached based on different analytical approaches [*Prévot and Camps*, 1993; *Love*, 1998, 2000].

[4] In this paper we report on a paleomagnetic record of the C27n-C26r GPR sampled in two separate lava sequences in the west Greenland flood basalt province. These data represent the ideal case of a volcanic GPR record with high temporal resolution. We will show that the two profiles record different and independent time series of the C27n-C26r GPR, which allow us to cast light on some of the questions that have arisen from single records.

2. Geology and Previous Results

[5] The onshore part of the west Greenland flood basalt province covers an area from Svartenhuk Halvø in north to the Disko Island in south of ~22,000 km^2 [*Clarke and Pedersen*, 1976] (Figures 1a and 1b). The main flood basalt sequence has an estimated stratigraphic thickness of 3–5 km [*Hald and Pedersen*, 1975], and it is divided in two main lithological units: Vaigat and Maligât Formations (Figure 1c). Our sampling was confined to the Vaigat Formation, which represents the earliest phase of volcanism and has a combined maximum thickness of ~2.1 km. An unusual large proportion of the Vaigat Formation eruptive units are picrites with a composition indicating an anomalously hot mantle producing high degrees of partial melting at relatively high pressures [*Holm et al.*, 1993; *Larsen and Pedersen*, 2000]. The lavas are typical P-type pahoehoe flows with smooth ropy surfaces indicating inflation on a subhorizontal paleosurface [*Self et al.*, 1998]. The volcanic productivity of the Vaigat Formation is estimated to have been as high as 1.2×10^{-3} km^3 yr^{-1} km^{-1} (rift) [*Pedersen et al.*, 2002], which is similar to the estimated 1.33×10^{-3} km^3 yr^{-1} km^{-1} for the SE Greenland 56–53 Ma seaward dipping reflector sequences [*Larsen and Saunders*, 1998] but an order of magnitude higher than the present-day Iceland mantle plume productivity, 1.3×10^{-4} km^3 yr^{-1} km^{-1}, in the neovolcanic rift zones of Iceland [*Pálmason*, 1982]. It is the extreme eruption rate, which makes large igneous provinces (like the North Atlantic one, Figure 1a) unique study sites for high-resolution paleomagnetic field recordings.

[6] The excellent geological exposure with large continuous sections cut by glacial erosion makes the west Greenland flood basalt province ideally suited for the geological application of multimodel photogrammetry [*Dueholm*, 1992], which has been a powerful tool for putting together the basalt stratigraphy and unraveling the highly dynamic volcanic history. On the basis of photogrammetry analyses, basin subsidence was estimated to ~25 m kyr^{-1} and volcanic aggradation to ~33 m kyr^{-1} during the C27n-C26r transition [*Pedersen et al.*, 2002]. Though the volcanism probably was of intermittent character, this very high aggradation rate suggests that the volcanic hiatuses must have been of relatively short duration. The only discernable volcanic hiatus, found just below Nuuk Killeq Member (Figure 1), is estimated to have lasted ~0.5 kyr [*Pedersen et al.*, 2002].

[7] Of particular interest is the ~17 km long photogrammetry section along the south coast of Nuussuaq peninsula [*Pedersen et al.*, 1993, 2002] which also covers the Nuusap Qaqqarsua profile (Figure 1b). In this section it is possible to trace individual Vaigat Formation lava flows over their entire length and in some cases observe subaerial lavas reach the paleoshoreline and laterally change into subaquous pillow lavas and hyaloclastite breccias. The individual flows vary in thickness from <1 to 10 m, and small lava domes with diameters <15 m are widespread. An observation of particular importance for the present study is that the lavas often thin out over short distances and rarely exceed 2 km before thinning out and disappearing.

[8] Previous magnetostratigraphy studies of the west Greenland flood basalts [*Riisager and Abrahamsen*, 1999] covered a composite thickness of ~1.6 km with one oriented hand sample from roughly every third flow. This study revealed two polarity zones, which based on ^{40}Ar/^{39}Ar geochronology [*Storey et al.*, 1998] were identified as C27n and C26r (Figure 1c). Moreover, a ~165 m thick sequence of intermediate directions recording the C27n-C26r GPR was found along the Nuussap Qaqqarsua profile (profile 1 on Figure 1b). Thellier paleointensity experiments were carried out on lavas from the pretransitional normal polarity zone, the zone of intermediate directions and the posttransitional reverse polarity zone [*Riisager and Abrahamsen*, 2000]. The asymmetric paleointensities around the C27n-C26r GPR were established, with an average C26r posttransitional field of 43.3 ± 15.0 μT (21 flows) significantly higher than the average C27n pretransitional intensity of 16.1 ± 6.9 μT (17 flows). The paleomagnetic field intensity during the GPR was as low as 7.4 μT.

3. Sampling

[9] For the present study we sampled 45 lava flows along the Nuussap Qaqqarsua profile and 35 flows in the Kuugannguup Innartaa profile (Figure 1b). The two profiles, distanced 16 km apart, can be correlated using a very distinctive brownish colored marker sequence of silica-enriched basaltic lavas (Nuuk Killeq Member; [*Pedersen*, 1985; *Pedersen et al.*, 1993]). Sampling along the Nuusap Qaqqarsua profile was concentrated in the zone where the magnetostratigraphy study of *Riisager and Abrahamsen* [1999] had previously identified the C27n-C26r GPR. The Kuugannguup Innartaa profile was sampled from the lowest subaerial flow up until just below the Nuuk Killeq Member.

[10] Paleomagnetic sampling was carried out on the accessible landward side of the mountains, where every exposed and accessible flow was sampled and their altitudes measured with an altimeter (listed in Table 1). Photogrammetry, on the other hand, was compiled on the near-vertical seaward side of the mountains where all lavas are clearly

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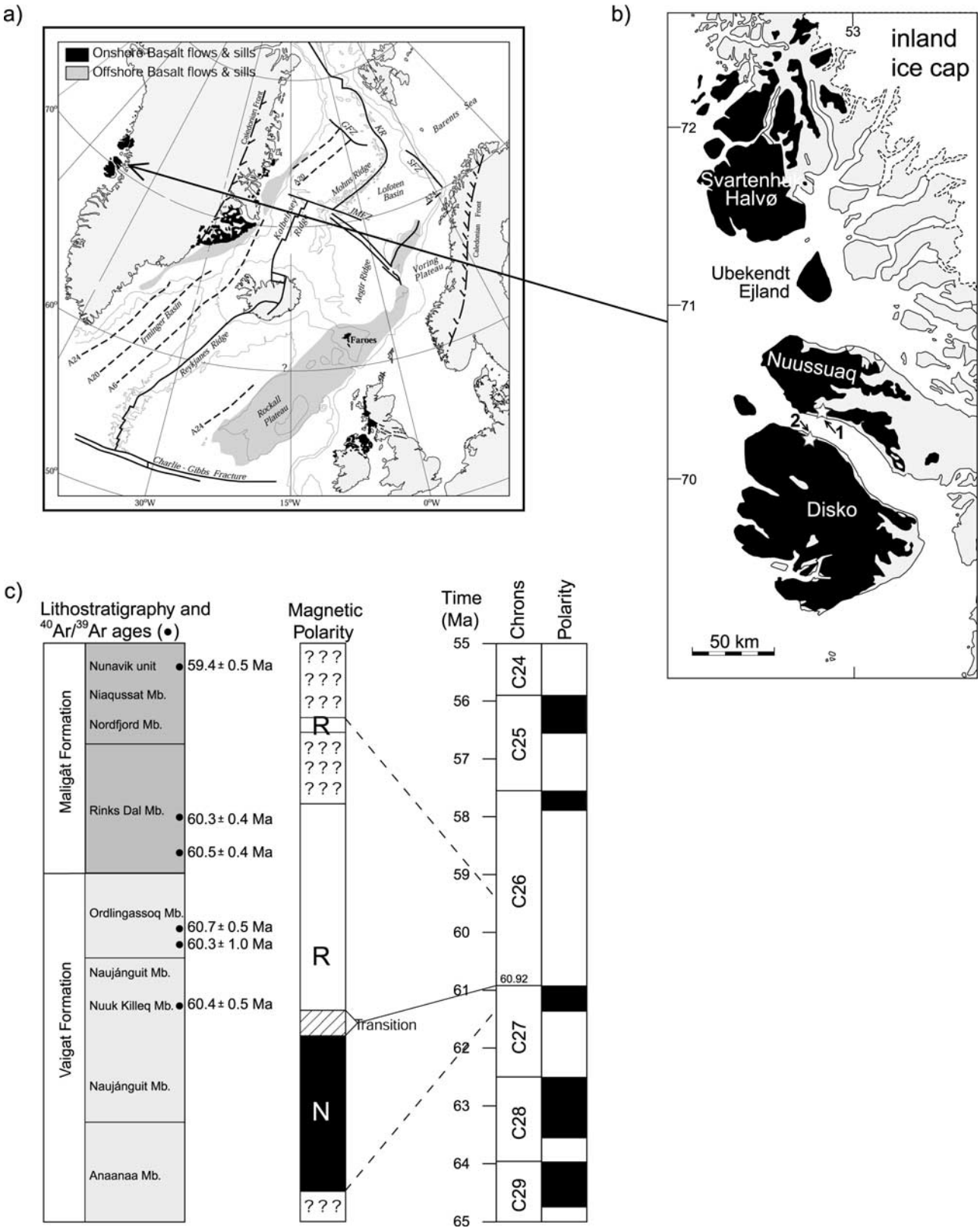


Figure 1. Maps of (a) the North Atlantic igneous province [after *Larsen and Saunders*, 1998]; (b) the Nuussuaq-Disko area in west Greenland, with the geographic position of the sampled sections shown by stars (1, the Nuusap Qaqqarsua profile; 2, the Kuugannguup Innartaa profile), and (c) lithostratigraphy of the west Greenland basalts and magnetostratigraphy of the Vaigat Formation after *Riisager and Abrahamsen* [1999] and *Riisager et al.* [1999]. The $^{40}Ar/^{39}Ar$ ages are from *Storey et al.* [1998]. Also shown is the geomagnetic polarity timescale of *Cande and Kent* [1995].

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Table 1. Paleomagnetic Results of the Two Sampled Profiles[a]

Flow	DG	Altitude, m	*D*, deg	*I*, deg	*N*	*k*	α_{95}, deg	VGP Lat, deg	VGPLong, deg
				Kuugannguup Innartaa					
1	1	230	−29.4	43.1	4	150	7.5	41.7	161.7
2	2	250	−26.3	51.0	4	1455	2.4	48.8	160.1
3	2	260	−25.2	47.8	4	238	6.0	46.3	157.7
4	2	265	−23.1	49.4	3	2445	2.5	48.0	155.6
5	2	275	−17.2	47.4	5	188	5.6	47.2	147.5
	Mean KI 2		−22.5	48.9	16	262	2.3	47.6	154.6
6	3	290	8.5	43.6	4	30	17.2	44.9	114.3
7	4	305	−12.5	63.1	5	441	3.7	63.6	145.3
8	5	330	−12.0	52.9	4	453	4.3	52.6	141.6
9	6	338	11.4	67.4	5	1438	2.0	69.3	104.2
10	6	348	10.5	64.5	8	253	3.5	65.6	107.4
11	6	352	7.7	67.1	4	395	4.6	69.2	111.0
12	6	360	13.8	64.7	4	1359	2.5	65.4	102.0
	Mean KI 6		10.8	65.7	21	416	1.5	67.1	106.3
13	7	380	245.9	47.3	5	299	4.4	19.1	247.0
14	8	395	246.8	60.3	3	973	3.9	31.4	251.2
15	8	408	243.8	58.4	3	293	7.2	28.5	252.7
	Mean KI 8		245.2	59.4	6	486	3.0	30.0	252.0
16	9	420	149.6	85.7	4	570	3.9	62.5	314.5
17	10	432	104.2	50.5	4	658	3.6	24.7	10.9
18	10	438	104.2	50.0	6	360	3.5	24.2	11.1
	Mean KI 10		104.3	50.2	10	487	2.2	24.4	10.9
19	11	450	108.4	41.8	3	558	5.2	16.7	9.9
20	12	460	35.7	60.7	6	984	2.1	56.2	73.6
21	13	472	43.8	31.4	3	651	4.8	30.5	74.9
22	14	475	90.9	−12.4	4	155	6.2	−6.2	36.4
23	14	480	91.3	−6.7	5	974	2.5	−3.6	35.0
24	14	498	91.2	−2.5	5	356	4.1	−1.6	34.4
25	14	500	89.2	−7.4	3	640	4.9	−3.2	37.1
26	14	502	94.7	−5.0	3	2200	2.6	−3.9	31.6
27	14	510	93.5	−0.1	3	481	5.6	−1.2	31.9
28	14	515	94.7	5.8	5	65	9.6	1.2	29.7
29	14	520	95.6	−1.3	3	1827	2.9	−2.5	30.1
30	14	521	95.2	−1.4	3	8284	1.4	−2.4	30.5
31	14	528	94.5	−3.8	3	214	8.5	−3.3	31.5
32	14	532	92.8	−0.4	3	510	5.5	−1.1	32.6
33	14	538	91.9	0.7	5	1127	2.3	−0.3	33.2
34	14	555	92.9	−5.2	3	257	7.7	−3.4	33.3
35	14	560	94.5	−5.8	3	497	5.5	−4.3	31.9
	Mean KI 14		92.9	−3.2	52	150	1.6	−2.5	32.9
				Nuusap Qaqqarsua					
1	1	840	56.6	49.4	4	993	2.9	39.4	56.3
2	1	850	58.8	48.0	4	181	6.9	37.5	54.6
3	1	860	50.8	47.0	4	725	3.4	39.2	63.4
	Mean NQ 1		55.3	48.2	12	315	2.5	38.8	58.1
4	2	865	−1.9	73.8	5	121	7.0	79.4	130.3
5	2	870	7.5	68.7	6	237	4.4	71.3	110.6
6	2	880	7.6	70.5	4	111	8.8	73.8	109.2
	Mean NQ 2		4.9	70.9	15	145	3.2	74.7	114.5
7	3	890	8.1	59.9	5	179	5.7	60.1	112.8
8	4	905	17.9	67.6	3	477	5.7	68.4	93.0
9	4	920	23.7	66.3	3	630	4.9	65.6	85.4
	Mean NQ 4		20.9	67.0	6	561	2.8	67.1	88.8
10	5	950	6.2	71.7	4	1008	2.9	75.8	111.0
11	6	965	52.2	39.2	4	126	8.2	33.1	64.3
12	7	970	46.2	51.2	4	102	9.1	44.0	66.7
13	7	980	48.8	56.8	5	354	4.1	48.4	61.0
	Mean NQ 7		47.5	54.3	9	153	4.2	46.4	63.8
14	8	985	31.3	4.2	7	1109	1.8	18.7	91.9
15	9	987	32.1	10.6	5	2251	1.3	21.7	90.4
16	10	990	30.9	4.0	7	480	2.8	18.7	92.3
17	11	1000	34.6	9.2	3	952	4.0	20.6	88.0
18	11	1005	33.5	10.9	3	248	7.9	21.6	88.9
	Mean NQ 11		34.1	10.1	6	452	3.2	21.1	88.4
19	12	1010	111.2	6.3	5	629	3.1	−4.0	14.0
20	13	1020	157.2	−12.2	4	177	6.9	−24.1	330.1
21	13	1022	156.6	−5.2	8	103	5.5	−20.5	330.2
22	13	1030	155.9	−7.1	5	44	11.7	−21.3	331.1
	Mean NQ 13		156.5	−7.4	17	82	3.9	−21.6	330.5
23	14	1045	1.1	−63.1	4	440	4.4	−25.1	124.3

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Table 1. (continued)

Flow	DG	Altitude, m	*D*, deg	*I*, deg	*N*	*k*	α_{95}, deg	VGP Lat, deg	VGPLong, deg
24	15	1048	13.3	−66.2	4	566	3.9	−29.4	115.1
25	16	1052	−5.7	−66.9	4	202	4.9	−30.0	129.4
26	17	1055	−9.6	−59.5	4	343	4.9	−21.0	133.0
27	17	1060	−5.7	−52.1	5	124	6.9	−13.2	130.1
	Mean NQ 17		−7.2	−55.4	9	141	4.5	−16.5	131.2
28	18	1062	−4.8	−65.6	5	250	4.9	−28.3	128.8
29	18	1070	−11.2	−63.9	7	310	3.4	−26.3	133.9
	Mean NQ 18		351.4	−64.7	12	276	2.6	−27.2	131.8
30	19	1078	4.7	−63.3	8	1081	1.7	−25.3	121.4
31	19	1080	3.6	−65.2	6	485	3.1	−27.7	122.4
32	19	1085	5.3	−66.0	4	718	3.4	−28.9	121.1
33	19	1090	4.3	−67.2	5	376	4.0	−30.4	121.9
	Mean NQ 19		4.4	−65.1	23	558	1.3	−27.6	121.8
34	20	1100	279.5	16.3	4	96	9.4	11.0	208.9
35	21	1105	211.2	−28.2	4	315	5.1	−31.4	269.3
36	22	1110	214.5	−13.5	3	360	6.5	−22.7	267.6
37	22	1113	217.4	−18.5	6	213	4.6	−24.7	263.9
38	22	1118	211.9	−19.7	5	171	5.8	−26.5	269.6
39	22	1125	213.2	−16.3	3	758	4.5	−24.4	268.6
40	22	1130	209.0	−15.6	5	678	2.9	−24.9	273.2
41	22	1140	214.4	−19.9	3	227	8.2	−26.1	266.9
	Mean NQ 22		213.4	−17.5	25	204	2.0	−25.0	268.3
42	23	1150	153.2	−72.4	4	632	3.7	−72.8	0.4
43	24	1160	132.4	−76.3	3	62	15.8	−71.1	39.1
44	25	1170	156.1	−65.9	4	167	7.1	−65.0	344.9
45	26	1178	151.9	−57.7	4	484	4.2	−54.7	344.9

[a]DG is for directional group; altitude is the altitude of the flow; *D*, *I* are the inclination and declination of the flow-mean ChRMs; *N* is number of samples used for calculation; *k*, α_{95} are the precision parameter and the confidence cone of the Fisher statistics [*Fisher et al.*, 1987]; VGP Lat, VGP Long are the latitude and longitude of the virtual geomagnetic pole. Mean directions for directional groups are also given. Geographic coordinates of the two profiles are 70.43°N, 305.13°E for the Nuusap Qaqqarsua profile and 70.25°N, 305.13°E for the Kuugannguup Innartaa profile.

visible. An exact flow-to-flow correlation between paleomagnetic and photogrammetry profiles is not possible, except for the distinctive Nuuk Killeq Member marker horizon where the altimeter readings agree well (within 10 m) with the exact altitudes obtained from photogrammetry. A comparison of the paleomagnetic and photogrammetric profiles reveals that roughly 30% of the flows were not sampled at the paleomagnetic profiles due to the limited exposure. Another advantage of the multimodel photogrammetry coverage is that individual lavas can be traced in three-dimensional space allowing very precise measurements (±0.5°) of the attitude of strata to be made. For the Kuugannguup Innartaa profile two lavas were traced over their length and found to dip 1.0° (strike ∼97°) and 2.3° (strike ∼38°), respectively. At the Nuusap Qaqqarsua profile the strike and dip of four lavas, from different altitudes, were measured and the dip varied between 1.1° and 2.8° with a NW strike in the lowest part and a SE strike higher in the profile. The small dip with varying strike directions most likely reflects minor local subsidence and/or paleotopography. Tectonic corrections are not applied in the further analyses.

[11] A total of 390 drill cores (three to seven cores per lava flow) were drilled directly in the field with a portable gasoline-powered drill. Drill cores were oriented using both magnetic and Sun compasses and the two readings agree well (within ∼7°). Because of the close proximity of the geomagnetic pole, and hence relatively small horizontal component of geomagnetic field at the sampling sites, magnetic compass readings are deemed less reliable and only Sun compass orientations are used in our analysis. In the laboratory the cores were subsampled into standard cylindrical specimens.

4. Results

4.1. Demagnetization Procedures

[12] Alternating field (AF) and thermal stepwise demagnetizations were performed in order to isolate the characteristic remanent magnetization (ChRM). The AF demagnetization was performed at the Geological Survey of Finland using a 2G cryogenic magnetometer with automatic three-axis AF demagnetizer. Thermal demagnetization was performed at Université Montpellier II with a laboratory-built furnace and a CTF cryogenic magnetometer. The ChRM directions were calculated using principal component analysis [*Kirschvink*, 1980].

[13] Roughly 25% of the measured specimens were subjected to thermal demagnetization (one to two specimens per site). All yielded essentially univectorial natural remanent magnetization demagnetizations, similar to the example shown in Figure 2a. The unblocking temperatures, ∼300–570°C, agree well with the main remanence carrier being oxy-exsolved titanomagnetite as inferred from previous rock magnetic and scanning electron microscopy observations [*Riisager and Abrahamsen*, 2000]. During standard three-axis AF demagnetization a majority of the samples acquired a strong gyroremanent magnetization (GRM) in AF fields above ∼60 mT (Figure 2b). All AF demagnetizations were therefore performed using the more laborious Zijderveld AF demagnetization technique [*Dankers and Zijderveld*, 1981; *Stephenson*, 1993] that

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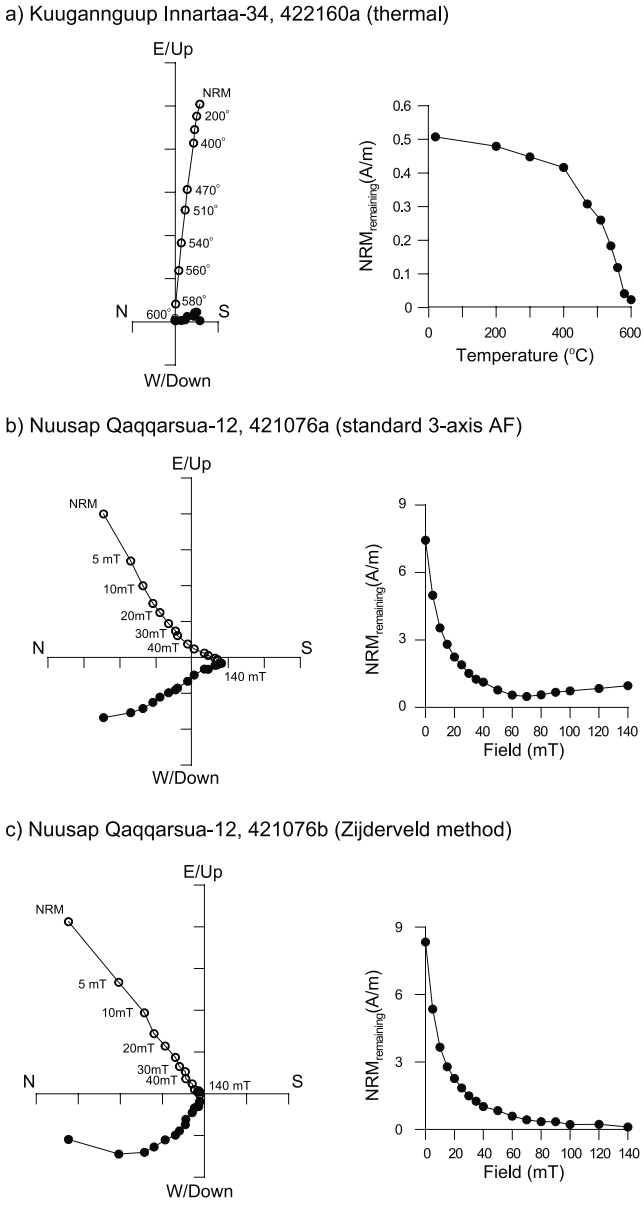


Figure 2. Orthogonal Zijderveld diagrams showing examples of (a) thermal demagnetization; (b) standard three-axis alternating field demagnetization showing a GRM acquisition at AF fields >60 mT; (c) AF demagnetization following Zijderveld method [*Dankers and Zijderveld*, 1981; *Stephenson*, 1993]. Note that Figures 2b and 2c are sister specimens. Solid/open circles represent projections on the vertical/horizontal plane. Sample numbers are GEUS numbers.

allows the GRM component to be subtracted from the NRM signal (compare Figures 2b and 2c). The ChRM directions obtained from the Zijderveld AF demagnetization technique are in excellent agreement with the ChRM directions defined from thermal demagnetization [*Riisager and Riisager*, 2001]. Further details regarding the rock magnetic properties and GRM acquisition will be subject of a separate manuscript.

[14] The direction of the ChRM component was defined, with maximum angular deviation <4°, for all 390 samples. This perfect success rate reflects the generally fresh nature of the Vaigat Formation picrites as well as the absence of lightning induced overprints, as thunderstorms presently do not occur in central west Greenland (~70°N).

4.2. Paleomagnetic Flow-Mean Directions and Directional Groups

[15] The flow-mean paleomagnetic directions are summarized in Table 1 and plotted versus stratigraphic heights in Figure 3. An obvious feature of both profiles is the presence of consecutive lavas having similar flow-mean directions (Figure 3). This feature is commonly observed in flood basalt provinces [*Riisager et al.*, 2002] and often taken to indicate that the series of flows were erupted in a very short period of time and therefore represent a spot reading of the paleomagnetic field [*Mankinen et al.*, 1985]. We must, however, stress that this interpretation is not straightforward unless supported by other evidences, as indistinguishable flow-mean directions also could reflect a standstill of the paleomagnetic field [*Hoffman*, 1992; *Love*, 1998, 2000]. We will return to this question in the discussion below. For the following analysis, directional groups (DGs) are defined as consecutive lava flows whose paleomagnetic directions are statistically indistinct at the 95% confidence level as determined by the Watson test [*Watson*, 1956]. From the altogether 80 sampled flows, we obtain 40 DGs, some of which consist of as many as 14 flows (KI 14, Table 1) others represent single flows. We note that bootstrap tests for common mean [*Tauxe*, 1998] yielded the exact same DGs. Unfortunately, neither photogrammetry nor geochemical analyses can be directly used to support this grouping because these data cannot be directly correlated to the paleomagnetic profiles (see section 3). These new results for the Nuussap Qaqqarsua profile cannot be flow-to-flow correlated with the magnetostratigraphic results of [*Riisager and Abrahamsen*, 1999] because the two sampling locations are different and different lava flows were sampled.

4.3. Stable Polarity Data and Paleomagnetic Pole

[16] We define a stable polarity DG as one having an angular distance between its mean VGP and the paleomagnetic pole smaller than a 40° cutoff angle [*Merrill and McFadden*, 1999]. Unfortunately, there exists no reliable paleomagnetic pole for Greenland at ~60 Ma, as previously published studies are based on either crude demagnetization techniques [*Tarling*, 1967; *Athavale and Sharma*, 1975] and/or dikes for which the tectonic corrections are highly uncertain [*Faller and Soper*, 1979; *Beckman*, 1982]. Instead, we use the 55–58 Ma Eurasian paleomagnetic pole from Faroe Islands [*Riisager et al.*, 2002], rotate it to Greenland with the rotation parameters of *Srivastava and Tapscott* [1986] and obtain pole coordinates latitude 68.1°N, longitude 162.1°E (A_{95} = 6.0°). Using this pole and the prechosen 40° cutoff angle we divide stable and transitional polarities. This division is largely insensitive to the chosen cutoff angle as any angle between 30° and 50° yields the same result. We obtain 14 DGs of stable polarity, where 10 DGs have normal polarity (KI 1–6, NQ 2–5; Table 1), and 4 DGs are of reversed polarity (NQ 23–26; Table 1). The corresponding paleomagnetic pole has coordinates: latitude 64.8°N, longitude 141.5°E (N = 14, K = 19.8, A_{95} = 9.2°).

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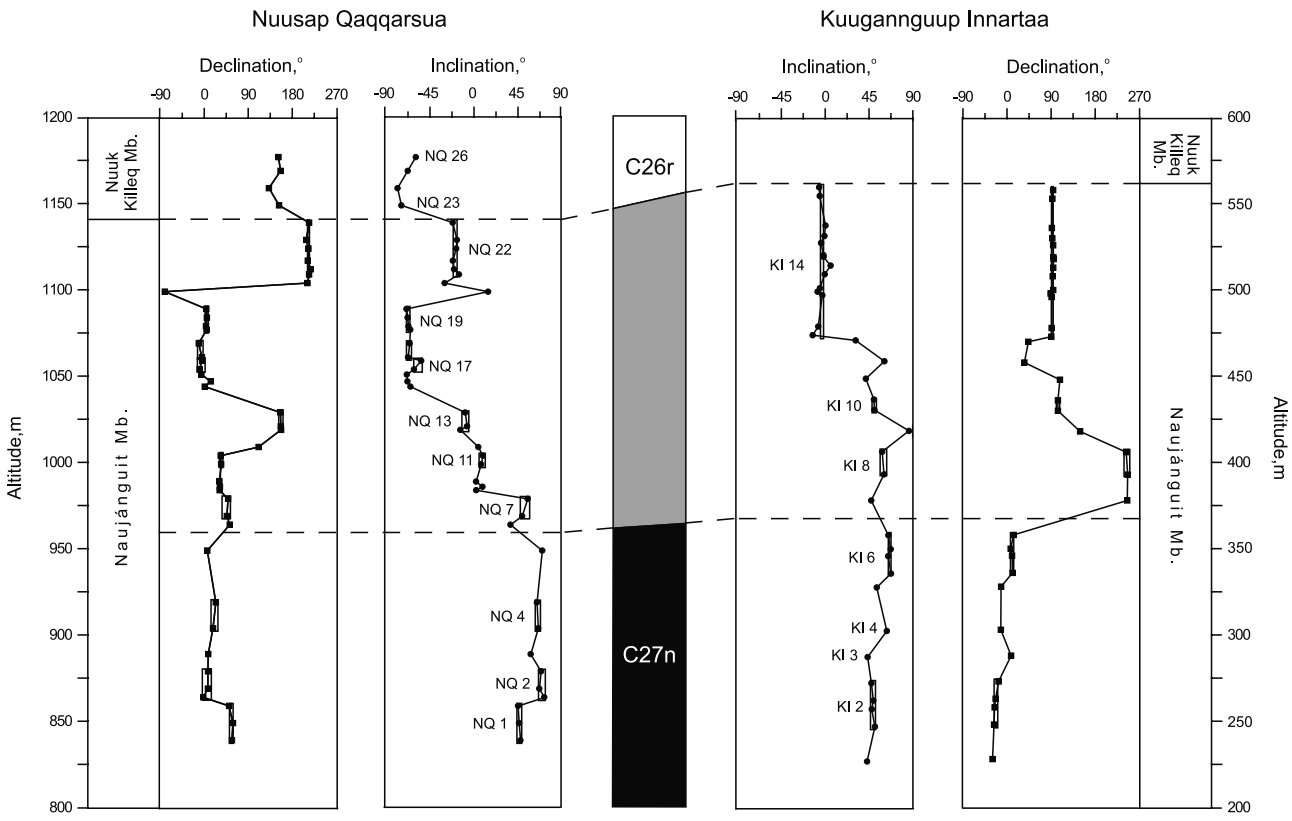


Figure 3. Flow-mean paleomagnetic directions of the Nuusap Qaqqarsua (NQ) and Kuuganngguup Innartaa (KI) profiles. Directional groups are shown with rectangles. Simplified lithostratigraphy is also shown. Dashed lines indicate zone of intermediate directions.

This pole is supported by a positive reversal test at classification C [*McFadden and McElhinny*, 1990]. A paleosecular variation estimate, expressed by the angular standard deviation (ASD) of the VGPs, is 18.3° with a 95% confidence limit of +6.7/−3.9° [*Cox*, 1969]. This ASD value is similar to the paleosecular variation estimate obtained from the contemporaneous Faroe Islands basalts at roughly the same paleolatitude (ASD = 21.7° +3.9°/−2.8° [*Riisager et al.*, 2002]). Finally, we note that our new 60 Ma paleomagnetic pole for Greenland is statistically indistinct at the 95% confidence level from the 55–58 Ma Faroes pole [*Riisager et al.*, 2002] rotated to Greenland.

4.4. Intermediate Polarity Data and the C27n-C26r Geomagnetic Polarity Reversal

[17] Following the definition of the above paragraph, we define the onset of the C27n-C26r GPR as the first lava with a VGP lying >40° away from the paleomagnetic pole. Altogether 23 flows in the Kuuganngguup Innartaa profile (DGs KI 7–14) and 31 flows in the Nuussap Qaqqarsua profile (DGs NQ 6–22) record transitional directions (Table 1 and Figure 3). In addition, NQ 1 consisting of three flows is cutoff, indicating a geomagnetic excursion in the weak C27n pretransitional field [*Riisager and Abrahamsen*, 2000]. The VGPs for all sampled flows and the corresponding DGs are given in Figures 4 and 5, respectively. In the Nuussap Qaqqarsua record (Figures 4a and 5a) after the initiation of the GPR the VGP crosses the equator first time on the west coast of Africa. Then through a rather rapid movement VGP clusters on Australia, crosses the equator second time and finally moves to the reverse polarity off eastern coast of South America. In the Kuuganngguup Innartaa profile (Figures 4b and 5b) VGP rapidly moves to North America, southern tip of Greenland, central Africa, then it returns close to the north pole before crossing equator on the east coast of Africa. Stippled line on Figures 4b and 5b signifies that the lava flows overlying KI 14, though not sampled, is the reversed polarity Nuuk Killeq Member (see also Figure 3).

[18] Although the significance of VGPs for intermediate field directions is not obvious, as the underlying assumption of a geocentric dipole field is probably flawed, the VGP paths are instructive for comparing the paleomagnetic directions. One observation is immediately clear from comparing Figures 4a and 4b, namely, that an exact flow-to-flow correlation between the two profiles is not possible. This is in accordance with the observation that a typical Vaigat Formation picrite lava flow has a mean length <2 km and hence much shorter than the 16 km distance between the two profiles. The VGP paths for the DGs (Figure 5) also differ significantly reflecting asynchronous volcanism with short volcanic hiatuses occurring at different times at the two profiles. General trends in the two records cannot be correlated unless one or both volcanic records contain longer lasting volcanic hiatuses of which there are no geological evidences (see section 2 and previous results). We therefore interpret the difference between the two records to indicate that the C27n-C26r transitional field

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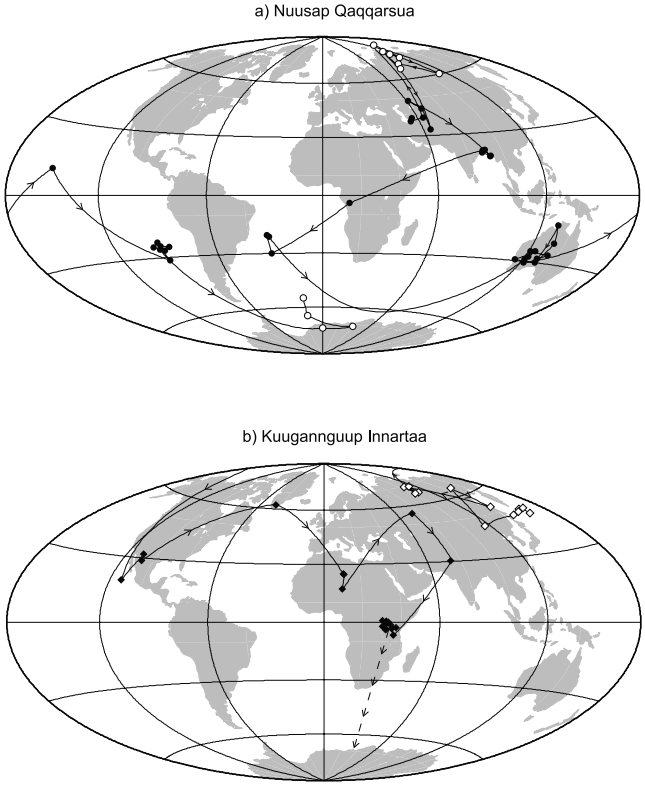


Figure 4. Hammer projection of the virtual geomagnetic poles for all lava flows of (a) the Nuusap Qaqqarsua profile and (b) the Kuugannguup Innartaa profile. Solid/open symbols are transitional/nontransitional directions. Continents are in the present geographic positions.

was more complex and faster changing than can be resolved by either record.

5. Discussions

5.1. The C27n-C26r GPR Record: What is the Significance of Directional Groups?

[19] Series of lava flows with similar paleomagnetic directions, like the DGs observed in the two C27n-C26r records (Figures 3–5), are also found in other high-resolution volcanic GPR records. Some authors interpret DGs to reflect a geomagnetic phenomenon, i.e., periods of time during which the field did not change [e.g., *Hoffman*, 1992; *Leonhardt et al.*, 2002], while others interpret them to be of volcanological origin, i.e., series of lavas erupted in a short period of time relative to the rate with which the field was changing [*Mankinen et al.*, 1985]. In the following we argue for a volcanological origin of the Vaigat Formation DGs suggesting that they reflect flow fields. First, we note that compound pahoehoe flow fields are believed to represent the dominant volcanic emplacement mechanism in continental flood basalt provinces [*Self et al.*, 1997], and since the Vaigat Formation lavas are very similar to typical pahoehoe flow fields of other volcanic provinces [*Self et al.*, 1998], they most likely were emplaced in a similar manner. On the basis of lava morphology, color, oxidation, and occurrence of soil horizons, as many as 17 flow fields have been defined in the GPR zone of the Nuusap Qaqqarsua profile comprising altogether 59 individual lavas (for further details, see *Pedersen et al.* [2002]). Similarities in the geochemical composition of successive Nuusap Qaqqarsua lavas further support the existence of flow fields and indicate a minimum of 6 in the GPR zone (L. M. Larsen and A. K. Pedersen, unpublished data, 2000). The strongest support for a volcanological origin of DGs comes from the fact that they cannot be directly correlated between the two paleomagnetic profiles. If we follow the reverse argument that DGs represent a more or less continuous recording of a standstill of the geomagnetic field, we would expect to find exact same DG directions in both profiles, which is not the case (Figure 5).

[20] It is very difficult to estimate the typical duration of a Vaigat Formation flow field. A detailed physical volcanology study of the five lava flows forming the ~14.7 Ma Roza Member compound pahoehoe flow field in the Colombia River flood basalt province suggests that it took ~14 years to be emplaced [*Thordarson and Self*, 1998]. The duration of Vaigat Formation flow fields may, however, have been different since the structure, composition and areal extent is different from the Roza Member. Given that the C27n-C26r GPR field was fast changing (see section 5.3) it is certainly possible that some of the flow fields had a duration long enough to record changes in the GPR field directions, i.e., one flow field could contain several statistically distinct DGs. While the 6 consecutive

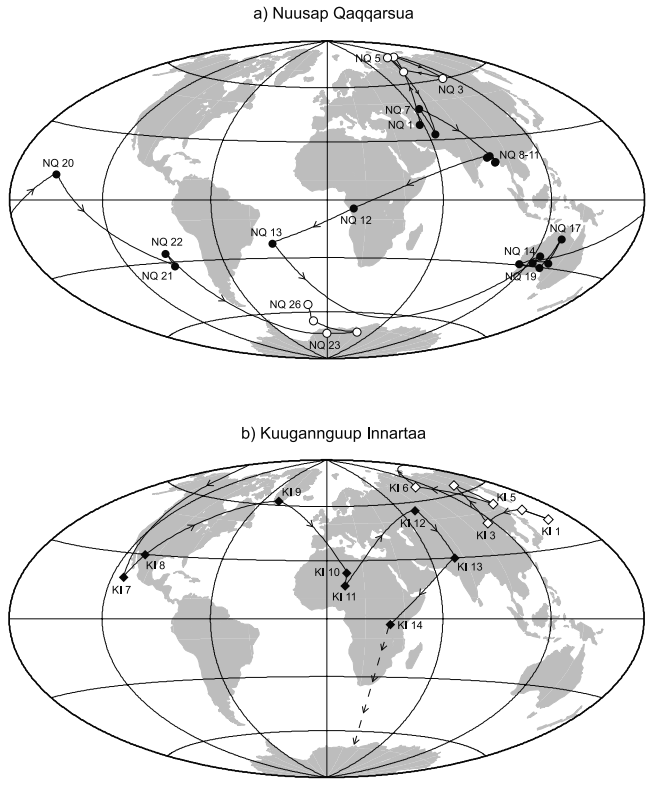


Figure 5. Hammer projection of the virtual geomagnetic poles for directional groups of (a) the Nuusap Qaqqarsua profile (NQ) and (b) the Kuugannguup Innartaa profile (KI). Numbers of directional groups are given (see also Table 1). Solid/open symbols are transitional/nontransitional directions. Continents are in the present geographic positions.

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DGs NQ 14–19 with VGPs over Australia (Figure 5a) could correspond to a preferred cluster suggested by *Hoffman* [1992], it could also represent a single long-lasting flow field as a similar grouping of VGPs is not observed in the Kuugannguup Innartaa profile (Figure 5b). The possibility that not even the DGs represent time-independent readings of the paleomagnetic field underlines the extreme difficulties in separating volcanological and geomagnetic events in volcanic GPR records.

5.2. The C27n-C26r GPR Record: Longitudinally Confined VGPs?

[21] Although longitudinal confinement cannot be proved nor disproved from the single GPR record presented here, several important observations can be made. First of all, we note that the apparent longitudinal confinement observed in the C27n-C26r lavas (Figure 6a) mainly reflects a volcanological artefact (i.e., DGs/flow fields), as several peaks disappear when DGs are considered (Figure 6b). An example is the pronounced peak between 30° and 60° for the Kuugannguup Innartaa profile (Figure 6a), which solely is caused by 14 flows recording the KI 14 DG. Second, the difference between the two C27n-C26r paleomagnetic records (Figures 4 and 5) also is reflected in the transitional longitudes being different (Figure 6). Although there are two apparent longitudinal bands at 60°–150° and 240°–270° (Figure 6b), which coincide with preferred longitudes observed in sedimentary records [*Laj et al.*, 1991], it is not clear if these bands are of geomagnetic or volcanological origin (see section 5.1). The distribution of longitudes (Figure 6b) is vastly different from, for example, the upper Olduvai sedimentary GPR record of *Tric et al.* [1991], where all transitional VGP longitudes (>100) fall in a very narrow band. This difference most likely relies on rock magnetic differences rather than differences between different GPRs.

5.3. The C27n-C26r GPR Record: How Complex is the Transitional Field?

[22] Is it possible that GPR fields are as complex and rapidly changing as suggested by recent numerical geodynamo models, with up to 1° changes in field direction per year [*Coe et al.*, 2000]? The two distinctly different records of the C27n-C26r GPR (Figures 4 and 5) clearly support this suggestion, as the transitional field certainly was faster changing than can be resolved by either profile, even though each of them represents some of the most detailed volcanic GPR records ever obtained. The different C27n-C26r GPR records also imply that the line connecting VGPs (Figures 4 and 5) is misleading as the true VGP path was much more complex than suggested by the connecting lines. Although very fast changing paleomagnetic field inescapably is smoothed in continuous sedimentary records, it is noteworthy that the highest resolution sedimentary GPR records also displays rather complex VGP paths [*Channell and Lehman*, 1997].

[23] Resolving possible yearly changes in the paleomagnetic field recorded in million year old geological records is obviously not a straightforward task and the true rate with which the transitional field is changing is presently unknown. Although it has been suggested that extremely rapid field changes were recorded within a single lava flow [*Coe et al.*, 1995], subsequent studies indicate that this most likely was a rock magnetic artefact [*Camps et al.*, 1999]. If flow fields and their approximate duration can be defined independently (through geochemical and physical volcanology studies), this would form the ideal tool for defining the true rate of GPR field changes.

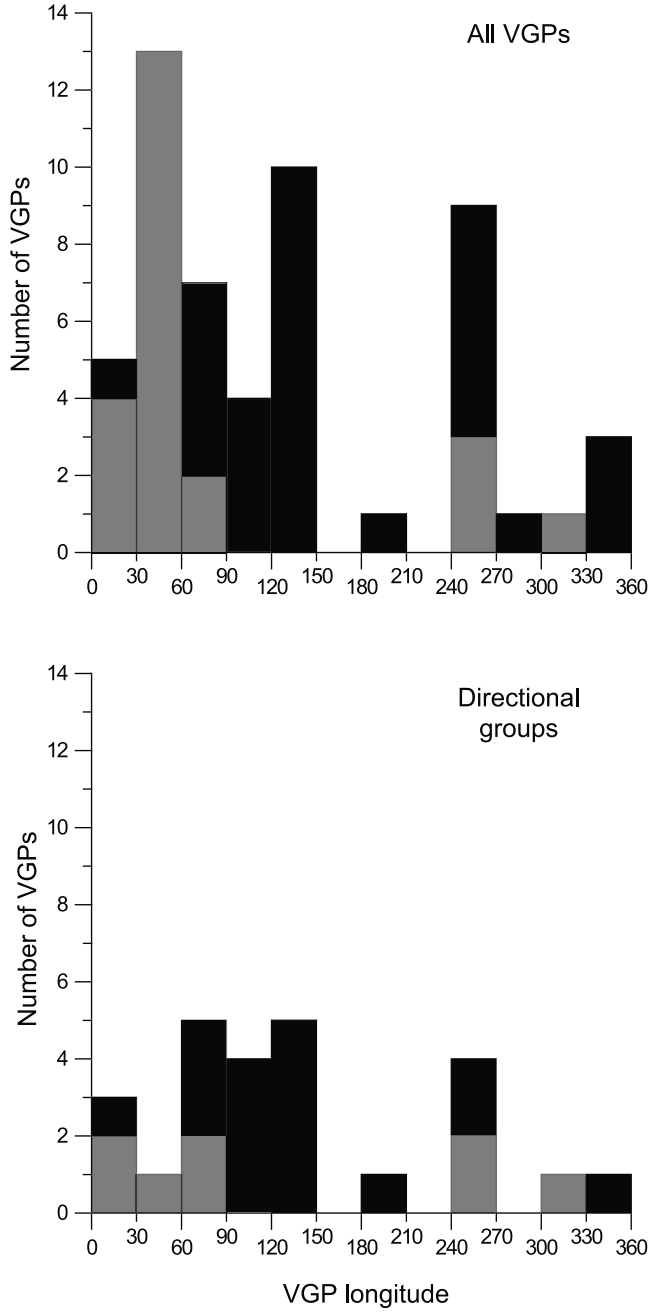


Figure 6. Histogram of stacked transitional virtual geomagnetic poles (VGPs) longitudes for (a) all VGPs and (b) directional group VGPs. Black is for the Nuussap Qaqqarsua profile, and gray is for the Kuugannguup Innartaa profile.

6. Conclusions

[24] The C27n-C26r GPR presented in this study (altogether 54 lavas of intermediate directions) owes its high

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temporal resolution to the unique geological setting of the west Greenland flood basalt province with continuous exposures of well-preserved lavas erupted at a very high eruption rate. Estimated thickness of the GPR zone in the Nuussap Qaqqarsua profile is ~165 m, and in the Kuuganngguup Innartaa it totals ~175 m. On the basis of photogrammetric, geochemical, and paleomagnetic evidences lava flows with statistically similar flow-mean paleomagnetic directions (DGs) are considered to represent flow fields. The importance of understanding the volcanological processes when studying GPRs is underlined. Failure to group flows into DGs is also critical for stable polarity data as it will lead to biased paleomagnetic poles and paleosecular variation estimates [see also *Riisager et al.*, 2002].

[25] The two high-resolution GPR paleomagnetic time series share no general trends in their VGP paths clearly indicating that the C27n-C26r transitional field was more complex and faster changing than can be resolved by either record. Analyses of the VGP longitudinal distribution indicate that apparent confinement observed in the C27n-C26r lava flows mainly reflects a volcanological artefact.

[26] Finally, the paleomagnetic pole (latitude 64.8°N, longitude 141.5°E, $N = 14$, $K = 19.8$, $A_{95} = 9.2°$) obtained in this study is the first pole from the Greenlandic part of the North Atlantic igneous province with positive field tests and based on detailed demagnetizaion experiments. This pole is in excellent agreement with the Faroes pole rotated to Greenland as well as paleomagnetic data from ongoing work on the overlying C26r lava flows of west Greenland.

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